higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N4

9 APRIL 2019

This marking guideline consists of 10 pages.

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✓ = 1 mark ✓ ½ = ½ mark

QUESTION 1

1.1 0 ; 1 ; 2 ; 3 ; 4 ; 5 ; 6 ; 7 (1)

1.2 $0{,}125_{10}$ ✓ (1)

1.3
$$\begin{array}{r} 100101_2 \\ 011010_2 \checkmark \\ + \quad\quad 1 \\ \hline 011011_2 \checkmark \end{array}$$
(2)

1.4 $\overline{AB} = \overline{A}+\overline{B}$ ✓

$\overline{A+B} = \overline{A}.\overline{B}$ ✓ (2)

1.5 1.5.1 $A\overline{A}+A=1$ ✓

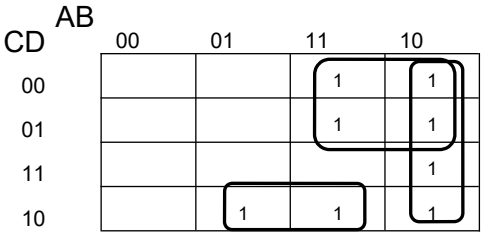
1.5.2 $A+ABCD+AC=A$ ✓

1.5.3 $A\overline{B}+A=A$ ✓

1.5.4 $A+B+\overline{A}=1$ ✓

(4 × 1) (4)

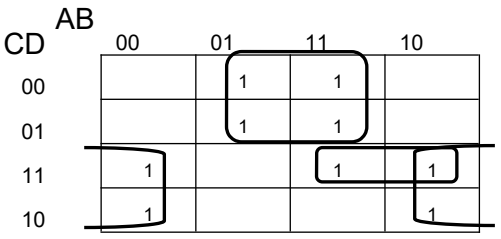
1.6 1.6.1



✓

$Z = A\overline{C}+A\overline{B}+B$ ✓

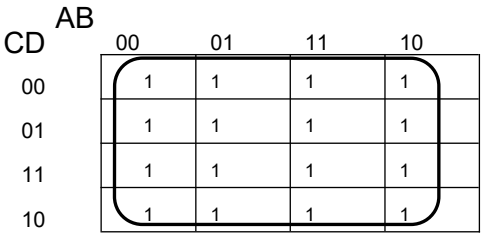
1.6.2



✓

$Z = B\overline{C}+ACD+\overline{B}C$ ✓

1.6.3



✓

$Z = 1$ ✓

(3 × 2) (6)

1.7

J	K	Q	$\overline{Q}$	
0	0	No change		✓
0	1	0	1	✓
1	0	1	0	✓
1	1	Toggle		✓

(4)
[20]

QUESTION 2

2.1 2.1.1 $122,3_8 - 132,C_{16}$

$122,3_8 = 001010010,011_2$ ✓½

$132,C_{16} = 000100110010,1100_2$ ✓½

$$\begin{array}{r} 001010010,011 \\ -\ 100110010,110 \\ \hline \end{array}$$

$$\begin{array}{r} 011001101,001 \\ +\qquad\qquad 1 \\ \hline 011001101,010 \end{array}$$ ✓

$$0c + 001010010,011$$ ✓

$100011111,101$ ✓

$$\begin{array}{r} 011100000,010 \\ +\qquad\qquad 1 \\ \hline -011100000,011 \end{array}$$ ✓

$-011100000,011_2 = -224,625_{10}$ ✓

2.1.2 $36,4_8 \times 6,25_{10}$

$36,4_8 = 011110,100_2$ ✓½

$6,25_{10} = 110,01_2$ ✓½

$$\begin{array}{r} 11110,100 \\ \times \quad 110,010 \\ \hline 111101000 \checkmark \\ 111101000000 \checkmark \\ + \quad 1111010000000 \checkmark \\ \hline 10111110,101000 \checkmark \end{array}$$

$1011111,101000 = 5F,A_{16}$ ✓

2.1.3 $54,6_8 \div 6,8_{16}$

$54,6_8 = 101100,110_2$ ✓½

$6,8_{16} = 0110,1000_2$ ✓½

$$\begin{array}{r} 110,111 \checkmark \\ 1101 \overline{)1011001,100} \\ \underline{-1101} \ \checkmark \\ 10010 \\ \underline{-1101} \ \checkmark \\ 10111 \\ \underline{-1101} \\ 10100 \\ \underline{-1101} \\ 1110 \\ \underline{-1101} \\ 1 \ \checkmark \end{array}$$

$110,111 = 6,7_8$ ✓

(3 × 6) (18)

2.2 $18,65_{10}$

$= 0001$ ✓ 1000✓ 0110✓ 0101✓

(4)

2.3 $0001011100011,0110_2$

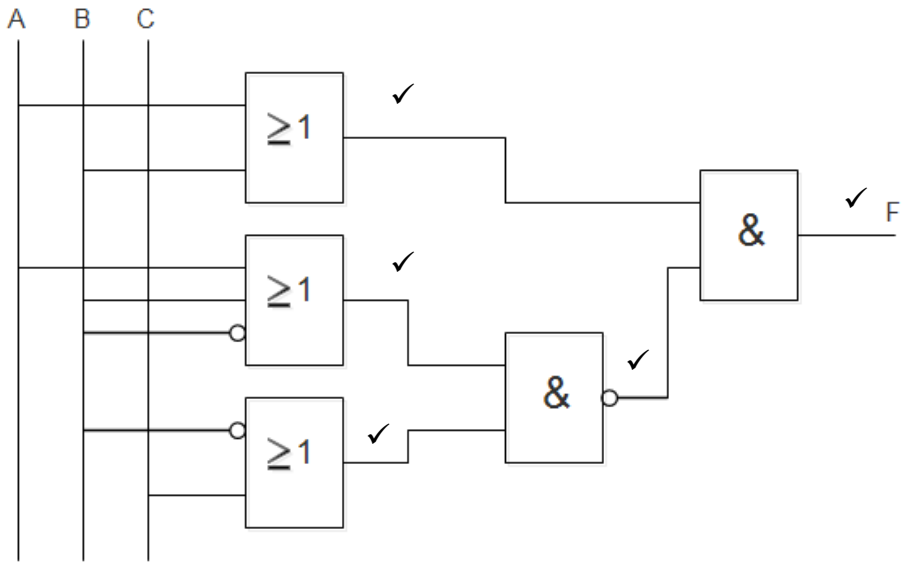
$=371,75_{10}$ ✓

$XS3 = 0110\ 1010\ \ 0100\ ,\ 1010\ 1000$ ✓✓

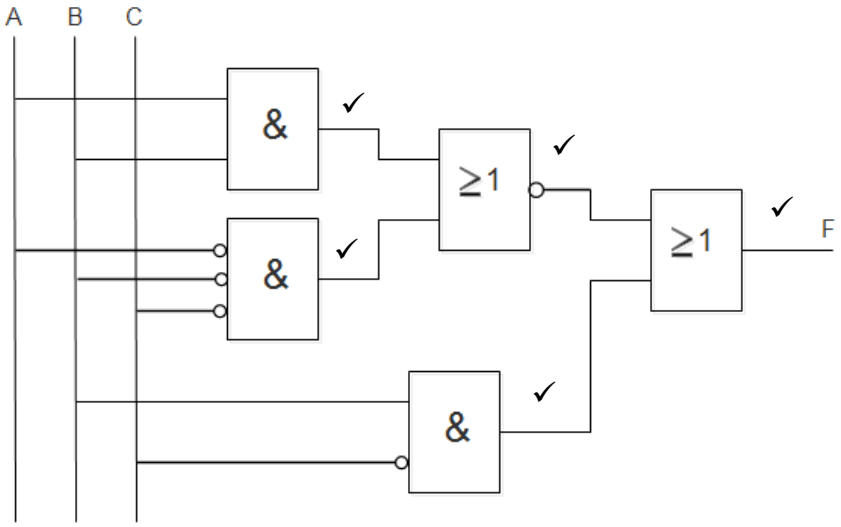
(3)
[25]

QUESTION 3

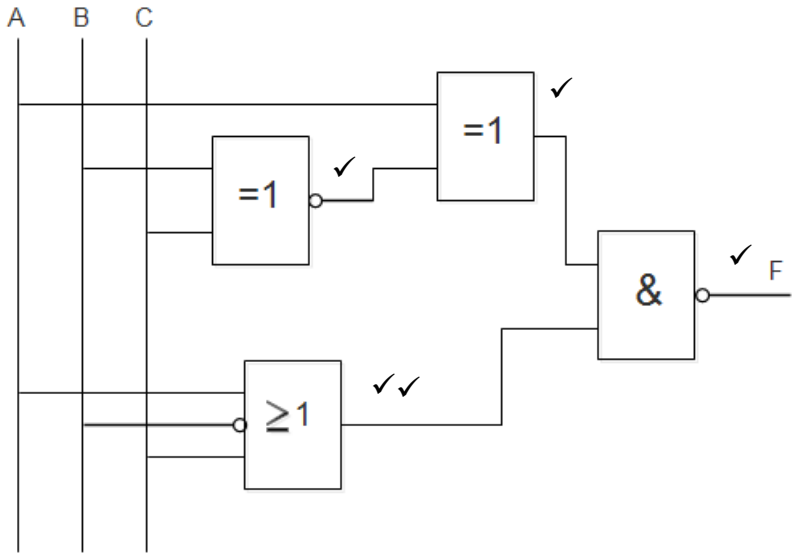
3.1 3.1.1



3.1.2



3.1.3



(3 × 5) (15)

3.2 3.2.1

$$F = (A+B)\,[\overline{A}(\overline{B}+\overline{C})] + \overline{A}C + A\overline{B}$$

$$= (A+B)(\overline{A}\overline{B}+\overline{A}\,\overline{C}) + \overline{A}C + A\overline{B} \quad \checkmark\tfrac{1}{2}$$

$$= 0 + 0 + 0 + \overline{A}B\,\overline{C} + \overline{A}C + A\overline{B} \quad \checkmark\tfrac{1}{2}$$

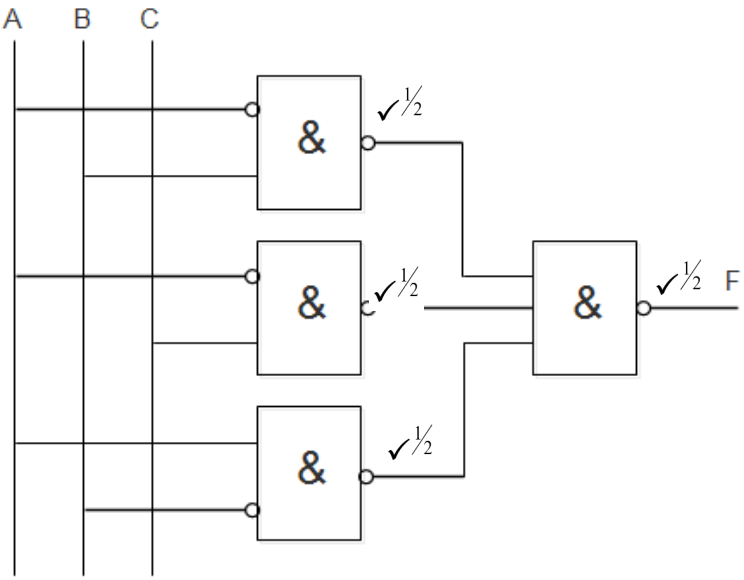
$$= \overline{A}(B\,\overline{C}+C) + A\overline{B} \quad \checkmark\tfrac{1}{2}$$

$$= \overline{A}(B+C) + A\overline{B}$$

$$= \overline{A}B + \overline{A}C + A\overline{B} \quad \checkmark\tfrac{1}{2}$$

$$= \overline{\overline{\overline{\overline{A}B + \overline{A}C + A\overline{B}}}} \quad \checkmark\tfrac{1}{2}$$

$$= \overline{\overline{A}B}\,.\,\overline{\overline{A}C}\,.\,\overline{A\overline{B}} \quad \checkmark\tfrac{1}{2}$$



3.2.2

$$F = \overline{\overline{(AB + \bar{A}BC)(A\bar{B}C + \bar{A}B\bar{C}) + (AB + A\bar{B}C)}}$$

$$= (AB + \bar{A}BC)(A\bar{B}C + \bar{A}B\bar{C}) + AB + A\bar{B}C \quad \checkmark^{1/2}$$

$$= AB + A\bar{B}C \quad \checkmark^{1/2}$$

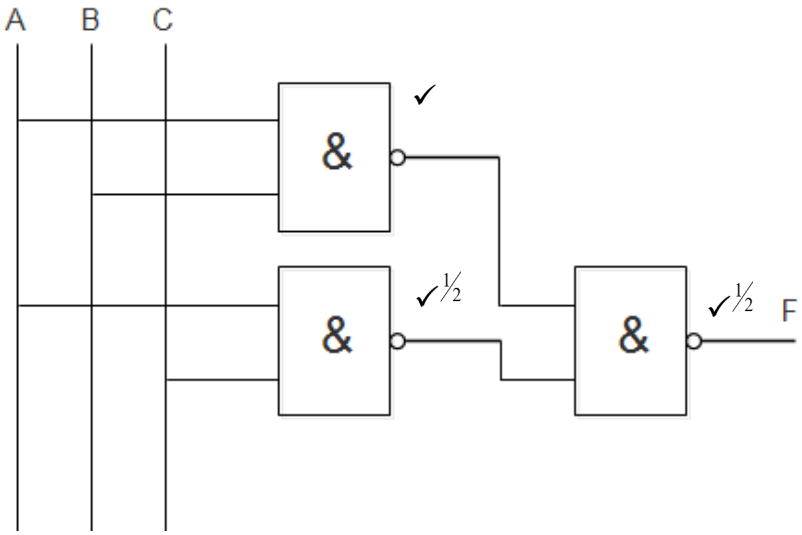
$$= A(B + \bar{B}C) \quad \checkmark^{1/2}$$

$$= A(B + C)$$

$$= AB + AC \quad \checkmark^{1/2}$$

$$= \overline{\overline{AB + AC}} \quad \checkmark^{1/2}$$

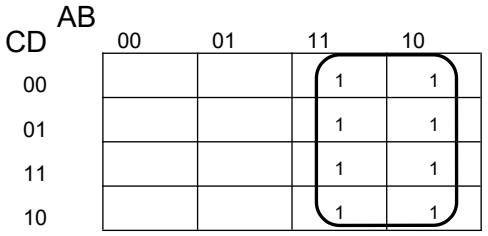
$$= \overline{\overline{AB}.\ \overline{AC}} \quad \checkmark^{1/2}$$



(2 × 5) (10)
[25]

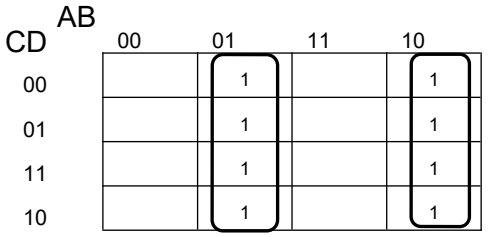
QUESTION 4

Input (Gray code)				Output (Binary)			
A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	1	0	1	0
✓	✓	✓	✓	✓	✓	✓	✓



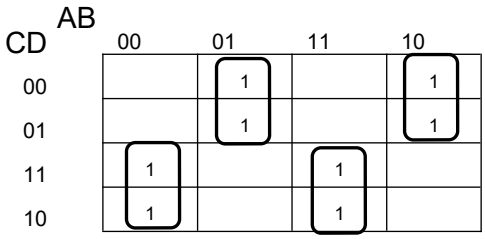
✓

W = A ✓



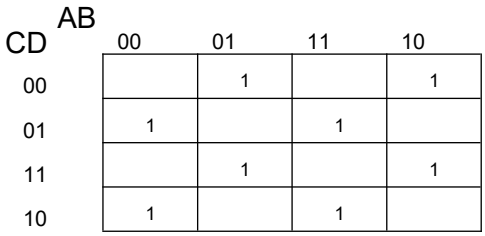
✓

$X = \bar{A}B + A\bar{B} = A \oplus B$ ✓



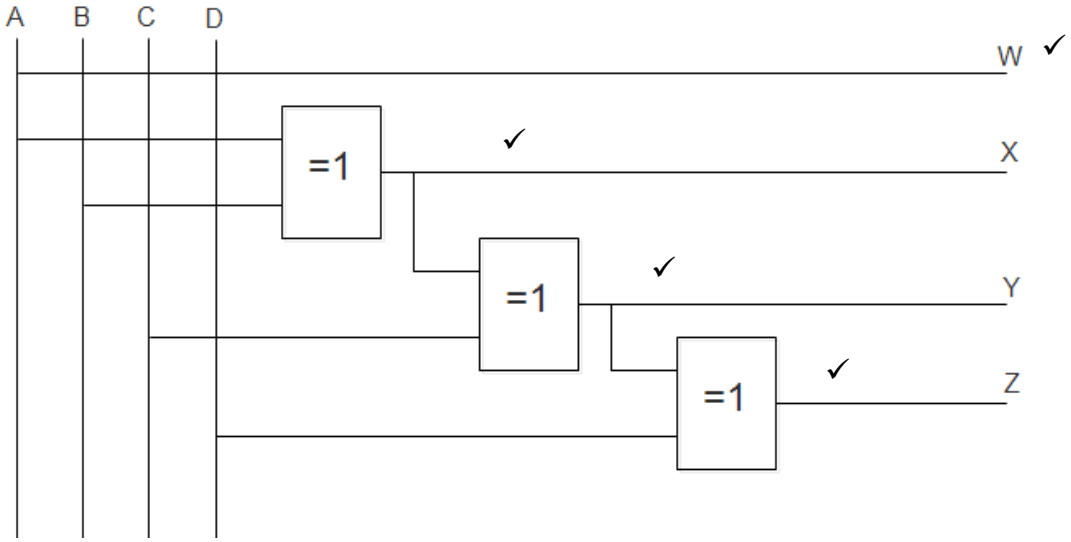
✓

$Y = A \oplus B \oplus C$ ✓



✓

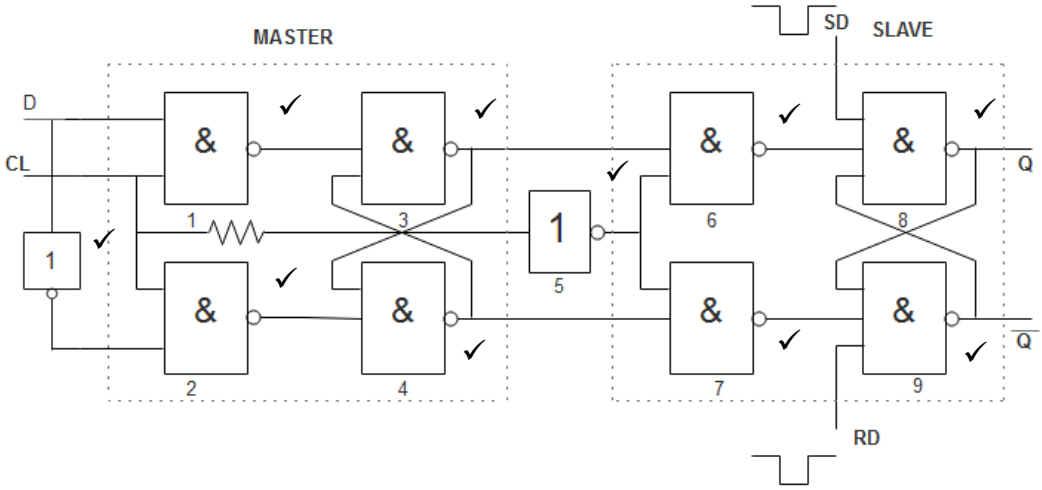
$Z = A \oplus B \oplus C \oplus D$ ✓



✓ ✓ ✓ ✓

[20]

QUESTION 5



[10]

TOTAL: 100